

Introduction to Radiation Effects on Devices

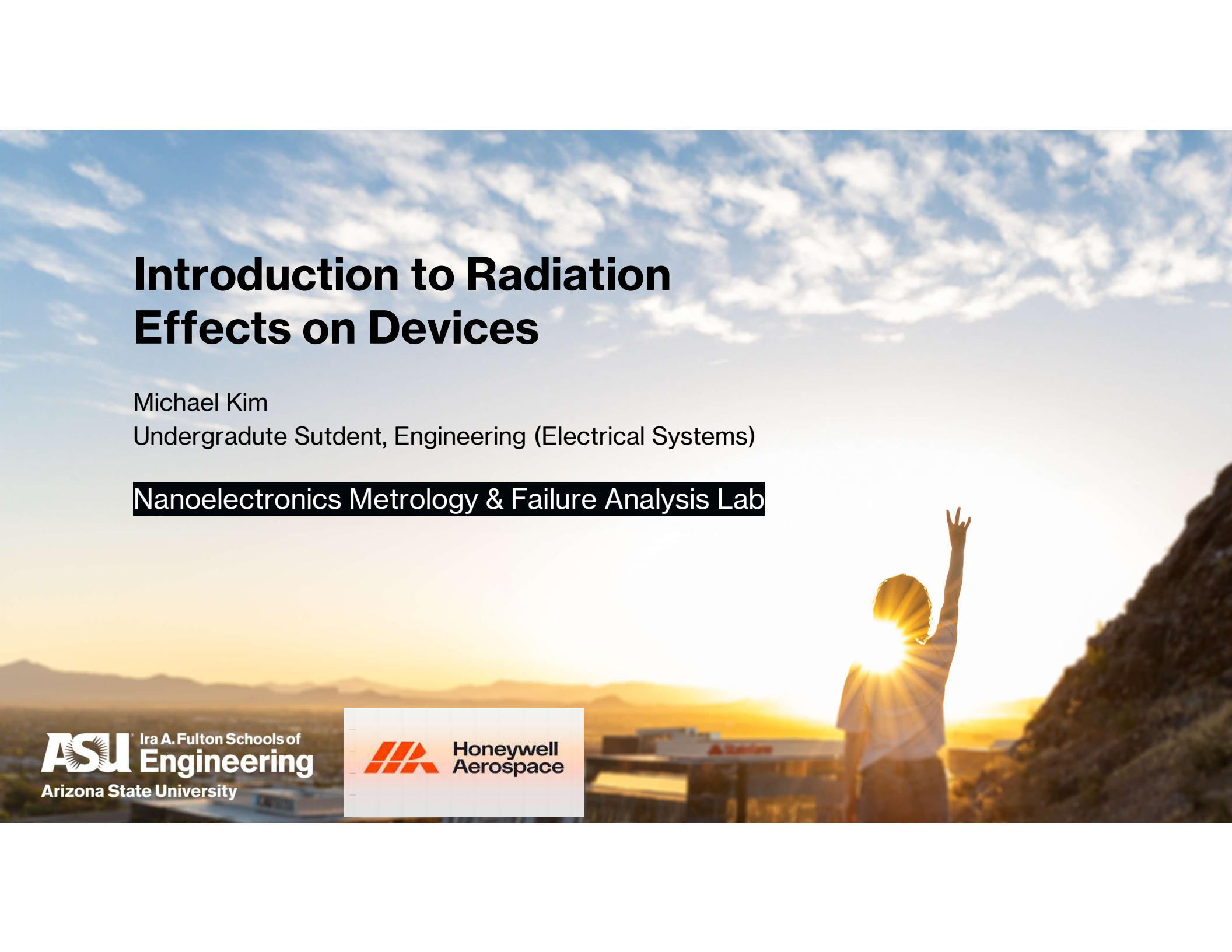
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Aerospace**



Outline



Introduction

Radiation Effects

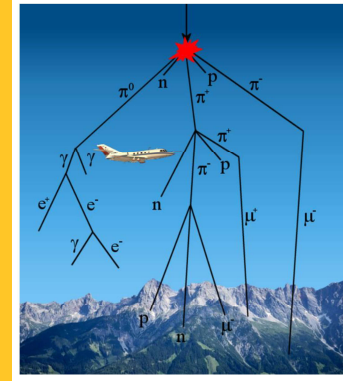
Effects on PN Junction

Diode response to Ionizing Radiation

Radiation Test of Diodes

Environments

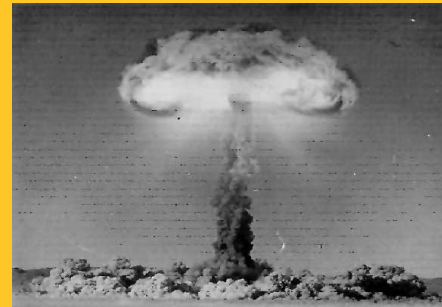
Radiation: Transfer of Energy by Photons and Particles.



Atmosphere



Natural Space



Weapon/Defense

Radiation Effects

Cumulative (Permanent)

- nDD
- TID

Transient (Temporary)

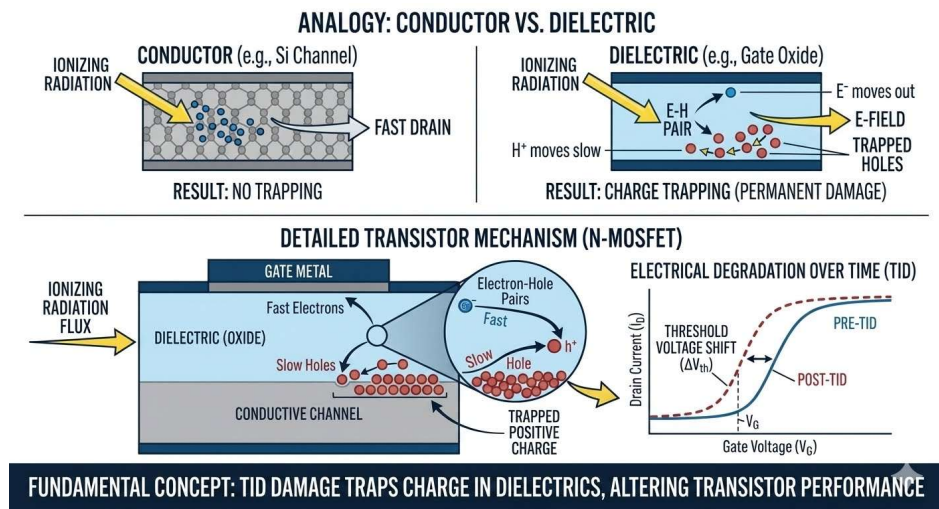
- Prompt Dose
- SEE
 - NDSEE
 1. SEU
 2. SET
 3. SEFI

Destructive

- DSEE
 1. SEB
 2. SEGR
 3. SEL
 4. SEDR

Total Ionizing Dose (TID)

- The degradation and eventual failure in semiconductor devices due to accumulated ionizing charges.
- For testing ,
 - HDR (High Dose Rate) $> 50 \text{ Rad(Si)/s}$
 - LDR (Low Dose Rate) $< 10 \text{ Rad(Si)/s}$
 - ELDRS (Enhanced Low Dose Rate Sensitivity) $< 10 \text{ mRad(Si)/s}$



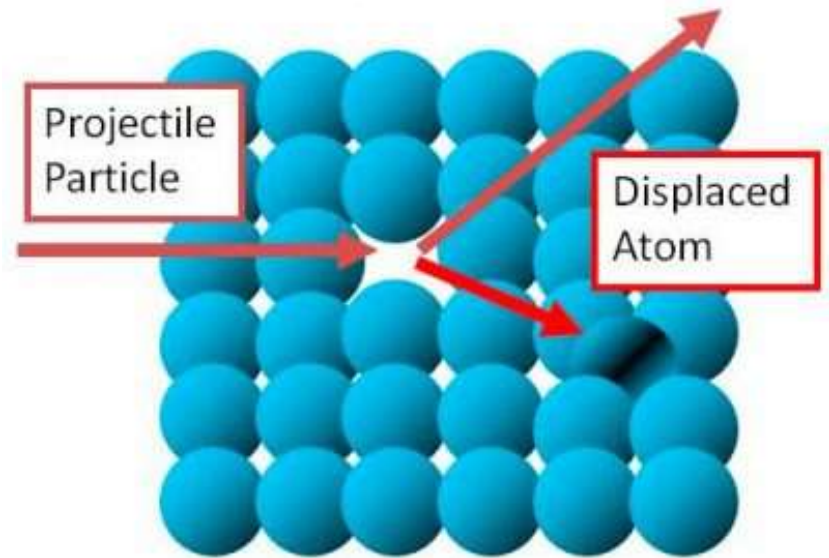
Unit of TID is rad or Gy:

$$1 \text{ rad}(si) = 0.01 \text{ Gy}(Si) = 0.01 \text{ J/kg}(Si)$$

Neutron Displacement Damage (nDD)

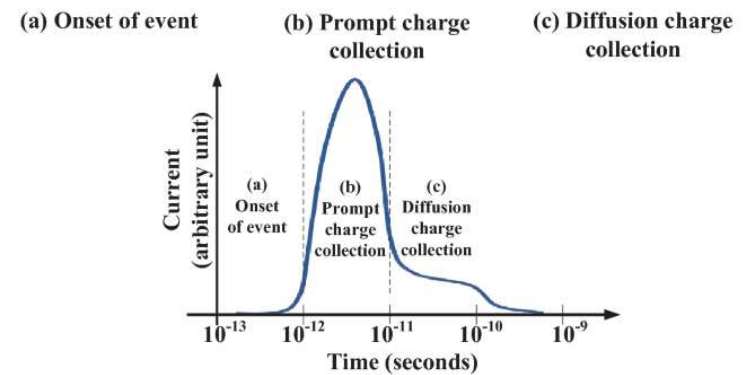
- High energy neutrons come in contact with the materials crystal lattice and physically knock out atoms from their normal position leaving a structural defect.
- Unit of nDD is Gy or rad, but for test purposes the parameter to change is the fluence:

n/cm^2 (1 MeV eq)



Prompt Dose

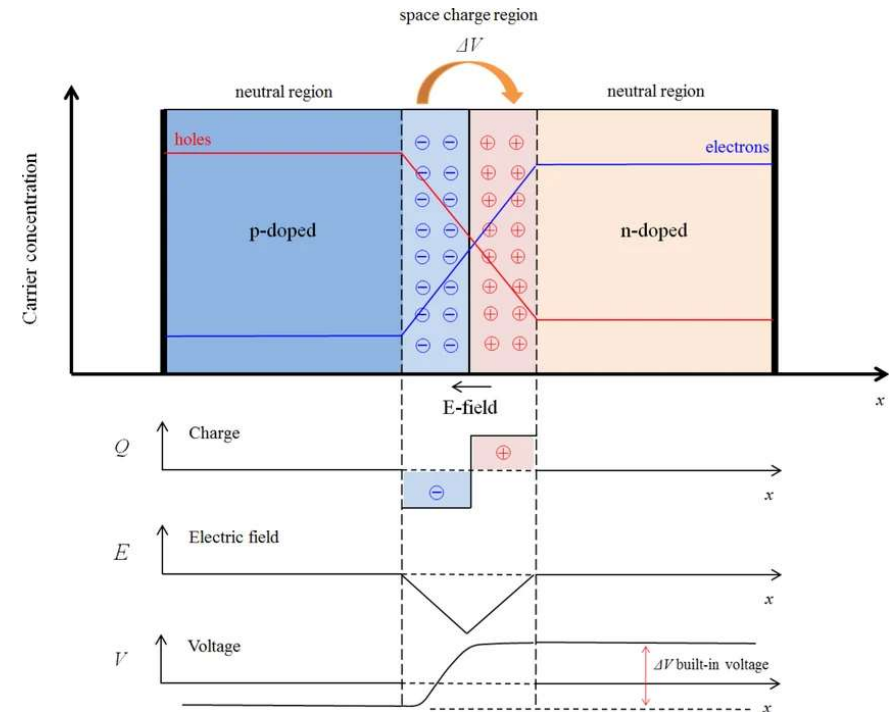
- When high energy particles pass through a biased diode, a large spike of current is observed.
 - Heavy Ions can create transients (GCRs)
 - Photons can create transients (X-Rays from manmade events)



Radiation Effects on Junction

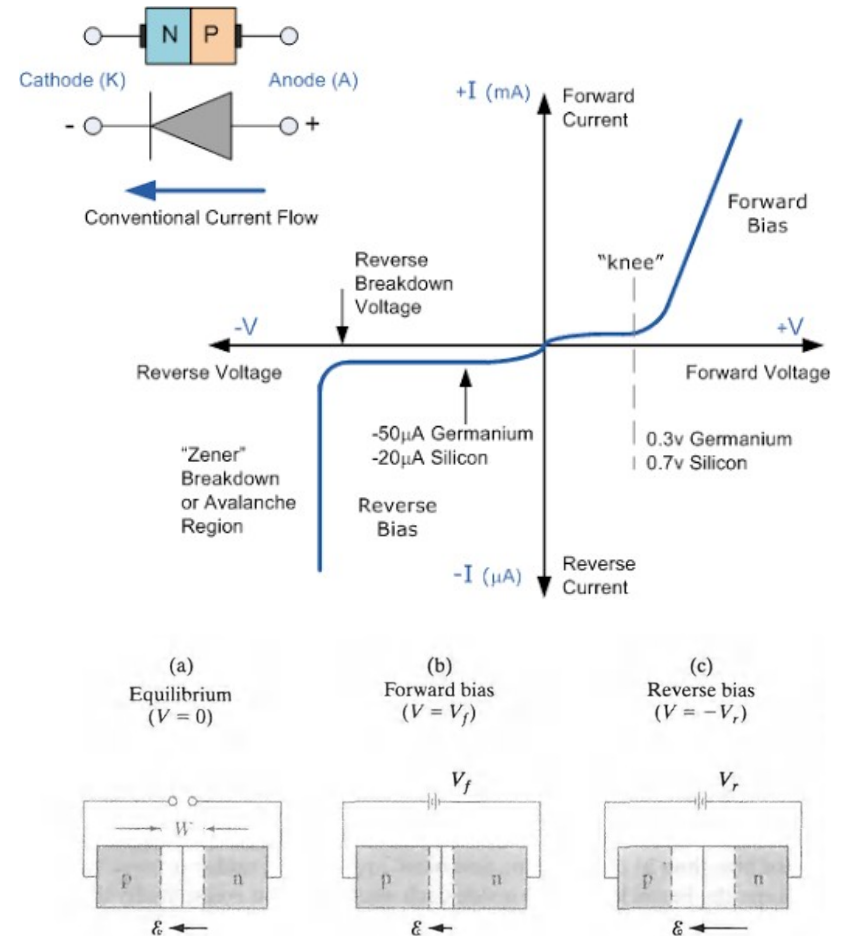
PN Junctions

- The **doping process** introduces intentional impurities, **creating free charge carriers (electrons or holes)** and donor/acceptor ions.
- A **Space Charge Region (SCR)** forms at the junction, where fixed ions create an internal electric field that prevents further charge movement.
- This Space Charge correlates to the built-in voltage or “**Threshold Voltage**”.



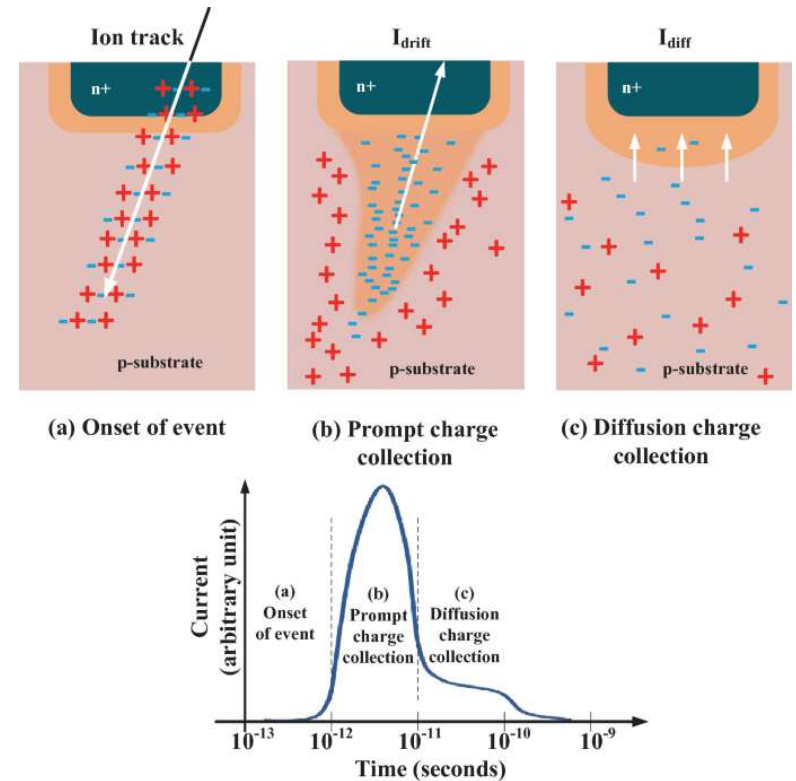
Diode – IV Characteristic

- The SCR is related to the width of the channel.
- Diodes are doped a specific way for different operation modes on the IV characteristic.
- For Radiation, the wider the width of SCR the more susceptible to radiation the junction is.



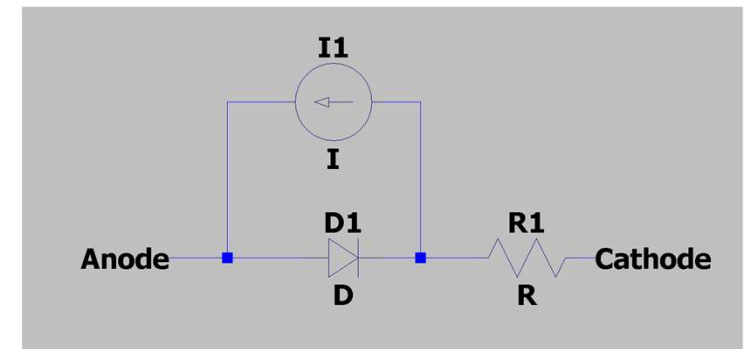
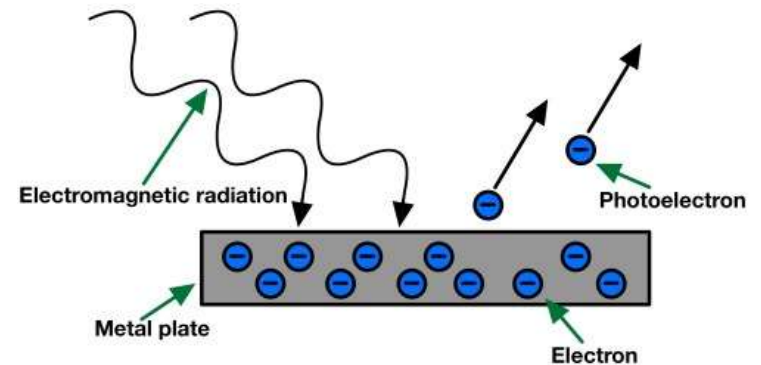
Radiation Response on PN Junction - Transients

- When Ionizing Radiation comes through electrons and holes are created.
- These electrons and holes can impact the size of the SCR.
- Electrons and holes separated by the E-Field in the SCR lead to parasitic currents.
- Once, electron and holes are collected the part and it's SCR return to initial conditions.



Radiation Effects - Transients

- PN Junctions behave the same way under radiation as photodiodes do to light.
- Particles/ Photons come through and ionize (create electrons/holes, ions).
- That Ionization creates a spike in current... which is harmful to electronics.



Equivalent circuit, where I1 represents the photocurrent waveform.

Thank You!

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Celano Lab